

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 13: Electricity and Magnetism — Lattice Friction and Stress Space

Version: 3.1 Date: July 6, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: C-311 Electric-Magnetic Duality, B-06b Four Views, A-04 Gradient, A-05b Pressure Response, E-02 Pressure Gradient Form, C-309 Friction Limit Status: GREEN (structure) / YELLOW (Maxwell derivation + stress-space formalization)

Gray — Standard Model Reference

Electricity and magnetism were unified by Maxwell in 1865. Maxwell's equations describe how electric and magnetic fields are produced and how they interact with each other and with charges. The four Maxwell equations: 1. $\nabla \cdot E = \rho/\epsilon_0$ (electric field from charge density) 2. $\nabla \cdot B = 0$ (no magnetic monopoles) 3. $\nabla \times E = -dB/dt$ (changing magnetic field produces electric field) 4. $\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 dE/dt$ (current and changing E produce B)

The electromagnetic force on a charge: $F = q(E + v \times B)$ (Lorentz force) Speed of light: $c = 1/\sqrt{\mu_0 \cdot \epsilon_0}$ (emerges from Maxwell's equations) Photons are electromagnetic waves — oscillating E and B fields propagating at c.

Standard Model strength: Maxwell's equations are exact for classical EM. QED extends this to quantum scale with extraordinary precision. Standard Model limitation: why the electron has charge e is not explained. Why magnetic monopoles do not exist is not deeply explained. The unification of EM with the weak force (electroweak) required a separate mechanism.

2D One-Wave Interpretation

In 2D, electricity and magnetism are not two separate phenomena.

They are the same pressure pattern in the lattice field viewed from two different angles.

In 2D: imagine a mode pressing outward in all directions from its boundary. That outward pressure — the radial component — is electricity. Now imagine that same mode spinning. The spinning creates a rotational pressure pattern around the mode. That rotational component is magnetism.

Same mode. Same pressure field. Two views of it.

The electric field is what you see when you look radially — in and out. The magnetic field is what you see when you look rotationally — around.

You cannot have one without the other. A moving charge always has both. A stationary charge has only the radial component (electric). A moving charge adds the rotational component (magnetic) from the motion.

3D One-Wave Interpretation

In 3D, electricity and magnetism are the radial and rotational components of one lattice pressure field P_{ψ} (C-311).

The boundary roll-off (B-06b) at a mode's boundary produces a pressure cushion P_c . That pressure cushion has two components:

Radial (in/out): $E_{\text{vec}} \sim \nabla P_c$ This is the electric field. It points away from positive charges (outward pressure cushion) and toward negative charges (inward pressure cushion).

Rotational (around): $B_{\text{vec}} \sim \nabla \times P_c$ This is the magnetic field. It circulates around the current direction. A moving charge has a rotational pressure component because the motion adds angular momentum to the cushion.

They are not two fields. They are one pressure cushion viewed from two angles. This is why Maxwell's equations couple E and B so tightly — they are the same quantity in different projections.

Electricity from lattice friction: When a mode moves through the lattice, it displaces the surrounding field. The displaced field resists the motion — this is resistance (γ). The resistance creates a pressure differential along the direction of motion. That pressure differential is the electromotive force (EMF) — voltage. Current is the flow of modes through the lattice under voltage. Resistance is the lattice's friction opposing that flow.

Ohm's law from lattice friction: $V = I * R$ In One-Wave: $V =$ pressure differential from mode displacement. $I =$ rate of mode flow. $R =$ lattice resistance (γ per unit length). Ohm's law is the One-Wave friction limit applied to current flow.

The magnetic shell around matter: As modes move through the lattice (current), they create a rotational pressure pattern. This rotational pressure is the magnetic field. The magnetic field circulates around the current — this is Ampere's law. A static charge has no magnetic field because there is no rotational pressure component. A moving charge has a magnetic field because the motion adds rotation.

The electric shell: The pressure cushion from the boundary roll-off (B-06b) locks the mode in place. It holds the charge in position within the lattice. The stronger the pressure cushion, the more firmly the charge is held. This is the binding energy of electrons in atoms.

Stress space and electricity: Electricity lives in stress space.

Stress space is the already-loaded pressure relation between coupled boundaries. It is not empty distance waiting for an object to travel across it. It is more like a straw already full of marbles.

If two marbles are pushed into one end of a filled straw, two marbles can pop out the other end with the same input pattern. The original marbles did not travel instantly from one end to the other. The pressure pattern moved through an already-filled, already-coupled path.

In electrical systems: Voltage = the push / pressure difference. Charge carriers = the marbles. The wire or field path = the filled stress path. The signal = the pressure pattern moving through the loaded line.

This is why electricity seems almost instant at human scale. The system is not waiting for matter to crawl across the whole distance. The stress condition updates through the coupled field, limited by the local friction limit.

In earlier language, this was called compressed space. The cleaner working term here is stress space: the compressed, already-coupled pressure field between boundaries. Full 1:6 compressed-space architecture is deferred.

Mathematics

Electric field from pressure cushion (C-311, B-06b): $E_{\text{vec}} \sim \nabla P_c = \nabla(\beta * \Delta E / V_c)$ E_{vec} points radially outward for positive charge (outward pressure cushion) E_{vec} points radially inward for negative charge (inward pressure cushion)

Magnetic field from rotational pressure (C-311): $B_{\text{vec}} \sim \nabla \times P_c$ For a current I in direction $\hat{z}$: B circulates around $\hat{z}$ — same as Biot-Savart law $|B| \sim I / r$ (at distance r from current)

Maxwell's equations from pressure field (sketch — Yellow):

1. $\nabla \cdot \mathbf{E} \sim \nabla \cdot (\nabla P_c) = \nabla^2 P_c \sim \rho/\epsilon_0$ (Poisson equation for pressure field)
2. $\nabla \times \mathbf{B} \sim \nabla \times (\nabla \times P_c) = 0$ (curl always divergence-free — no magnetic monopoles)
3. $\nabla \times \mathbf{E} \sim -dB/dt$ (changing rotational pressure creates radial pressure change — Faraday)
4. $\nabla \times \mathbf{B} \sim \mu_0 \mathbf{J} + \mu_0 \epsilon_0 d\mathbf{E}/dt$ (current creates rotational pressure — Ampere-Maxwell)

All four Maxwell equations emerge from the single pressure field P_c . Formal derivation deferred.

Speed of light from pressure field: $c = 1/\sqrt{\mu_0 * \epsilon_0}$ In One-Wave: $c = v_{\max} = \text{friction limit} = \sqrt{\beta_{\max}} * \Delta x / \Delta t$ Identification: $\mu_0 * \epsilon_0 = 1/\beta_{\max} * (\Delta t/\Delta x)^2$ Formal derivation deferred.

Ohm's law: $V = \Delta P$ (pressure differential along direction of motion) $I = n_{\text{modes}} * v_{\text{drift}} * A$ (mode density * drift velocity * cross-section) $R = \gamma * L / A$ (lattice resistance * length / area) $V = I * R$ follows from $V = \gamma * (I * L / A) / 1 = I * R$. QED.

Stress-space transfer: Let $S(x,t)$ be the stress state along a coupled path. An input pattern changes the local stress:

$$\Delta S_{\text{in}}(t) = S(0,t) - S_0$$

The output responds after propagation through the coupled path:

$$\Delta S_{\text{out}}(t + \tau) \sim \Delta S_{\text{in}}(t) * A_{\text{loss}}$$

where: $\tau = L / v_{\text{stress}}$ $v_{\text{stress}} \leq v_{\max}$ $A_{\text{loss}} = \text{attenuation from resistance / friction}$

In the ideal low-loss case: $A_{\text{loss}} \rightarrow 1$ ΔS_{out} preserves the input pattern.

Marble-straw analogy: The individual carriers do not need to traverse the full distance instantly. The stress pattern transfers through the already-filled path.

Predictions

1. Maxwell's equations emerge from the single pressure field P_c . Prediction: all four Maxwell equations follow from ∇P_c and $\nabla \times P_c$. No separate electric and magnetic fields needed. One pressure field, two projections.
2. No magnetic monopoles exist. Prediction: $\nabla \times \mathbf{B} = \nabla \times (\nabla \times P_c) = 0$ exactly. Magnetic monopoles would require a source for the rotational pressure — but rotational pressure is always sourceless (curl is always divergence-free).
3. The speed of light equals the friction limit. Prediction: $c = 1/\sqrt{\mu_0 \epsilon_0} = v_{\max}$ from the pressure field propagation speed. Formal identification of μ_0 and ϵ_0 with lattice parameters deferred.
4. Ohm's law is the lattice friction law applied to current flow. Prediction: $R = \gamma * L / A$ where γ is the lattice resistance coefficient. Different materials have different γ values — this is the One-Wave basis of resistivity.
5. Superconductivity occurs when $\gamma \rightarrow 0$ locally. Prediction: superconductivity is the condition where lattice resistance drops to zero. In One-Wave: this happens when mode coupling becomes perfectly coherent — all modes oscillate in phase, no energy lost to out-of-phase collisions. The Cooper pair is two modes in perfect phase synchronization (B-15 Hyperloop at electron scale).
6. Electrical communication is stress-pattern transfer through an already-loaded path. Prediction: signal propagation is not carrier-object travel across the full distance. It is stress update through a coupled field or conductor. Signal speed depends on the local stress propagation limit, resistance, and coupling geometry.

Yellow Audit

- Maxwell equations derivation from pressure field sketch level only
 - Formal identification of μ_0 , ϵ_0 with lattice parameters deferred
 - Superconductivity from $\gamma \rightarrow 0$ is structural claim — derivation deferred
 - Cooper pair as Hyperloop at electron scale needs formalization
 - Ohm's law derivation is sketch level — formal proof deferred
 - Stress-space transfer model needs formal propagation equation
 - Marble-straw analogy is explanatory only, not final math
 - Lorentz force derivation from pressure field not yet done
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Future Work

Derive all four Maxwell equations formally from $P_c = \beta * \Delta E / V_c$. Identify μ_0 and ϵ_0 with lattice parameters. Derive Lorentz force from pressure field gradient. Model superconductivity as $\gamma \rightarrow 0$ phase coherence. Formalize Cooper pair as electron-scale Hyperloop. Derive Ohm's law rigorously from lattice friction. Formalize stress-space transfer as $\Delta S_{in} \rightarrow \Delta S_{out}$ through coupled paths. Separate conductor stress-transfer from field/radio stress-transfer.

Closing Thoughts

Maxwell unified electricity and magnetism in 1865. One-Wave goes one step further: both are the same pressure field. Not two unified fields. One field with two projection angles.

The electric field is what you see when you look radially at the pressure cushion. The magnetic field is what you see when you look rotationally. A moving charge shows both because motion adds rotation to the cushion.

Electricity is not mysterious. It is stress-space transfer.

A wire is not an empty tunnel waiting for energy marbles to sprint from one end to the other. It is more like a straw already full of marbles. Push a pattern into one end, and the matching pattern appears at the other end because the whole path is already filled and coupled.

When modes flow through the lattice against resistance (γ), the pressure differential is voltage and the flow is current. Ohm's law is the lattice's friction law.

Magnetism is the rotation of that same pressure. When charges move, they rotate their pressure cushions. That rotation is what other charges and currents respond to.

One pressure field. Two views. One stress-space transfer path. All of electromagnetism.

END OF BOOK 1 CHAPTER 13 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.